



DPG School of Technology & Management

Sample Question Paper

Subject : Computer Graphics

Subject Code : 24BCA404DS04

Course : BCA

Academic Year : 2025-2026

Doc ID : 3cc12890-a49f-4aae-9335-94e572286d38

Paper Code : CG-633380

Section A (5 × 2 = 10)

1. What is a graphics pipeline?
 2. What is a viewport?
 3. What is transformation composition?
 4. What is hidden surface removal?
 5. Define Gouraud shading.
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Section B (Long Questions)

Unit 1

1. A graphics system renders 60 frames per second. Calculate the number of frames rendered in 5 seconds and discuss performance. (10 Marks)
 2. Explain Computer Graphics and its applications in detail. (10 Marks)
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Unit 2

1. Describe the complete windowing process with diagram. (10 Marks)
 2. Explain windowing and viewport concepts. (10 Marks)
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Unit 3

1. Discuss projection techniques with suitable diagrams. (10 Marks)
 2. Find orthographic projection of points A(2,3,4) and B(5,6,7) on XY plane. (10 Marks)
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Unit 4

1. Explain Phong Reflection Model in detail. (10 Marks)
 2. Compute diffuse reflection component using suitable numerical example. (10 Marks)
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Academic Year : 2025-2026

Doc ID : 0a6bb668-cc6b-4d00-a519-c2765a62bb02

Paper Code : CG-685510

Section A (5 × 2 = 10)

1. What is a vertex?
2. Define Bresenham line algorithm.
3. Translate point (1,2,3) by (2,3,4).
4. If mipmap level reduces size from 512 × 512 to half, find next level size.
5. What is a clipping window?

Section B (Long Questions)

Unit 1

1. Explain the graphics pipeline and rendering process. (10 Marks)
2. Describe the historical development of Computer Graphics. (10 Marks)

Unit 2

1. Explain DDA line drawing algorithm with example. (10 Marks)
2. Explain Bresenham's line drawing algorithm with example. (10 Marks)

Unit 3

1. Solve numerical problems based on 3D transformations. (10 Marks)
2. Differentiate Orthographic and Perspective Projection. (10 Marks)

Unit 4

1. Explain mipmapping with advantages and limitations. (10 Marks)
2. Differentiate Gouraud and Phong Shading techniques. (10 Marks)





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Doc ID : 2ce096e9-0e9d-45fc-b448-99fc371d3f4d

Paper Code : CG-519283

Section A (5 × 2 = 10)

1. What is raster graphics?
2. What is HSL color model?
3. What is 3D transformation?
4. Define illumination.
5. What is Midpoint Subdivision method?

Section B (Long Questions)

Unit 1

1. Explain Computer Graphics and its applications in detail. (10 Marks)
2. Explain graphics primitives and their importance. (10 Marks)

Unit 2

1. Explain DDA line drawing algorithm with example. (10 Marks)
2. Explain Bresenham's line drawing algorithm with example. (10 Marks)

Unit 3

1. Explain 3D Translation, Rotation and Scaling with matrices. (10 Marks)
2. Clip a polygon against rectangular window using Sutherland-Hodgman method. (10 Marks)

Unit 4

1. Explain texture mapping and texture coordinates. (10 Marks)
2. Discuss ambient, diffuse and specular reflections. (10 Marks)





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Doc ID : 169c604b-8664-40db-ad7c-d4c0e9cdbdf5

Paper Code : CG-766248

Section A ($5 \times 2 = 10$)

1. Find the total pixels in an image of size 800×600 .
2. Define Bresenham line algorithm.
3. Define Sutherland-Hodgman algorithm.
4. Define Gouraud shading.
5. Calculate diffuse component when $K_d=0.5$ and light intensity=0.8.

Section B (Long Questions)

Unit 1

1. Describe the historical development of Computer Graphics. (10 Marks)
2. Discuss the stages of graphics pipeline with suitable examples. (10 Marks)

Unit 2

1. Explain 2D Translation, Rotation and Scaling with matrices. (10 Marks)
2. Explain composite transformations in detail. (10 Marks)

Unit 3

1. Apply Cohen-Sutherland clipping algorithm on line joining (2,2) and (12,8). (10 Marks)
2. Differentiate Orthographic and Perspective Projection. (10 Marks)

Unit 4

1. Describe image rendering and texture mapping process. (10 Marks)
2. Using Phong reflection model, calculate ambient reflection for given values. (10 Marks)





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Subject Code : 24BCA404DS04

Course : BCA

Academic Year : 2025-2026

Doc ID : e878d158-41be-4fbf-b6de-44cc5adec15d

Paper Code : CG-875761

Section A ($5 \times 2 = 10$)

1. Name two applications of Computer Graphics.
2. What is viewport transformation?
3. What is line clipping?
4. What is depth buffering?
5. What is 3D scaling?

Section B (Long Questions)

Unit 1

1. Explain Computer Graphics and its applications in detail. (10 Marks)
2. Discuss Cartesian and Homogeneous coordinate systems in detail. (10 Marks)

Unit 2

1. Rotate point (2,0) by 90 degrees about origin. (10 Marks)
2. Perform translation of triangle A(1,1), B(2,1), C(1,2) by (4,3). (10 Marks)

Unit 3

1. Explain point, line and polygon clipping techniques. (10 Marks)
2. Differentiate Orthographic and Perspective Projection. (10 Marks)

Unit 4

1. Explain visibility determination in Computer Graphics. (10 Marks)
2. Explain Phong Reflection Model in detail. (10 Marks)





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Subject Code : 24BCA404DS04

Course : BCA

Academic Year : 2025-2026

Doc ID : eda10190-8a72-4f89-9df3-c101fed02c19

Paper Code : CG-269725

Section A ($5 \times 2 = 10$)

1. What is a transformation matrix?
2. What is interpolation?
3. Define clipping.
4. Define illumination.
5. Calculate diffuse component when $K_d=0.5$ and light intensity=0.8.

Section B (Long Questions)

Unit 1

1. Explain the graphics pipeline and rendering process. (10 Marks)
2. Explain graphics primitives and their importance. (10 Marks)

Unit 2

1. Explain region codes and their significance. (10 Marks)
2. Discuss reflection and its types with examples. (10 Marks)

Unit 3

1. Apply scaling transformation on cube vertices using scale factor 2. (10 Marks)
2. Describe Sutherland-Hodgman polygon clipping algorithm. (10 Marks)

Unit 4

1. Using Phong reflection model, calculate ambient reflection for given values. (10 Marks)
2. Explain mipmapping with advantages and limitations. (10 Marks)





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Course : BCA

Academic Year : 2025-2026

Doc ID : 089bc5dd-5913-4d95-916e-04ba3237f1b7

Paper Code : CG-255199

Section A (5 × 2 = 10)

1. Find the total pixels in an image of size 800 × 600.
 2. What is a window?
 3. Define Sutherland-Hodgman algorithm.
 4. What is visibility determination?
 5. What is rotation?
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Section B (Long Questions)

Unit 1

1. An image has resolution 1024 × 768. Calculate the total number of pixels and explain its significance. (10 Marks)
 2. Explain Computer Graphics and its applications in detail. (10 Marks)
-

Unit 2

1. Explain windowing and viewport concepts. (10 Marks)
 2. Apply Bresenham line drawing algorithm for line from (0,0) to (6,4). (10 Marks)
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Unit 3

1. Discuss homogeneous coordinates in 3D graphics. (10 Marks)
 2. Describe Sutherland-Hodgman polygon clipping algorithm. (10 Marks)
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Unit 4

1. Explain illumination models used in Computer Graphics. (10 Marks)
 2. Compute diffuse reflection component using suitable numerical example. (10 Marks)
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Academic Year : 2025-2026

Doc ID : 4458e2ac-eb2c-4528-ae8a-9633995173e6

Paper Code : CG-898058

Section A (5 × 2 = 10)

1. What is a graphics pipeline?
2. Using DDA algorithm, find the first two points for line from (0,0) to (4,4).
3. What is Midpoint Subdivision method?
4. Define Phong shading.
5. Define DDA algorithm.

Section B (Long Questions)

Unit 1

1. Explain transformation matrices and their applications. (10 Marks)
2. Discuss the stages of graphics pipeline with suitable examples. (10 Marks)

Unit 2

1. Explain DDA line drawing algorithm with example. (10 Marks)
2. Explain windowing and viewport concepts. (10 Marks)

Unit 3

1. Solve numerical problems based on 3D transformations. (10 Marks)
2. Explain Perspective Projection and Vanishing Points. (10 Marks)

Unit 4

1. Explain mipmapping with advantages and limitations. (10 Marks)
2. Discuss Scan Line Algorithm for hidden surface removal. (10 Marks)





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Academic Year : 2025-2026

Doc ID : ebcc5086-46c2-466e-9e29-064722e5eff5

Paper Code : CG-557369

Section A ($5 \times 2 = 10$)

1. What is a vertex?
2. What are region codes?
3. What is 3D transformation?
4. What is depth buffering?
5. What is a projection plane?

Section B (Long Questions)

Unit 1

1. Discuss the stages of graphics pipeline with suitable examples. (10 Marks)
2. Calculate the distance between points A(2,3) and B(8,11) using distance formula. (10 Marks)

Unit 2

1. Explain region codes and their significance. (10 Marks)
2. Explain Bresenham's line drawing algorithm with example. (10 Marks)

Unit 3

1. Clip a polygon against rectangular window using Sutherland-Hodgman method. (10 Marks)
2. Explain 3D Translation, Rotation and Scaling with matrices. (10 Marks)

Unit 4

1. Compare different hidden surface removal techniques. (10 Marks)
2. Discuss ambient, diffuse and specular reflections. (10 Marks)





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Sample Question Paper

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Course : BCA

Academic Year : 2025-2026

Doc ID : 384d7e26-91ba-4676-afde-bbe78294e3e7

Paper Code : CG-777579

Section A (5 × 2 = 10)

1. What is a vertex?
2. What is HSL color model?
3. What is transformation composition?
4. A scene contains 100 polygons. If 40 are hidden, how many remain visible?
5. What is transformation composition?

Section B (Long Questions)

Unit 1

1. Explain graphics primitives and their importance. (10 Marks)
2. A graphics system renders 60 frames per second. Calculate the number of frames rendered in 5 seconds and discuss performance. (10 Marks)

Unit 2

1. Explain DDA line drawing algorithm with example. (10 Marks)
2. Perform translation of triangle A(1,1), B(2,1), C(1,2) by (4,3). (10 Marks)

Unit 3

1. Explain composite 3D transformations with examples. (10 Marks)
2. Discuss homogeneous coordinates in 3D graphics. (10 Marks)

Unit 4

1. Discuss Scan Line Algorithm for hidden surface removal. (10 Marks)
2. Using Phong reflection model, calculate ambient reflection for given values. (10 Marks)

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